Voyager 1 Is Almost One Light-Day Away

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NASA's Voyager 1 spacecraft is close to an amazing **milestone**. Soon, it will be one **light-day** from Earth. That means even light the fastest thing there is needs a whole day to travel from the spacecraft to us.

Voyager 1 left Earth in 1977, almost 50 years ago. It is the farthest spacecraft ever built, and it has flown into the space between the stars.

The spacecraft is running low on power. It loses about 4 **watts** of power every year. So in September, engineers finished a careful project. They replaced three devices with new ones that use less electricity. This should keep Voyager 1's science tools working longer.

In October, the spacecraft's signal suddenly stopped for about two days, then came back on its own. One set of small engines had stopped working, and the spacecraft had switched to **backup** engines by itself. "We have to expect the unexpected," said mission manager Kareem Badaruddin.

Word Treasure

light-day -- The distance light travels in one whole day -- used to measure huge distances in space.

milestone -- An important point or achievement that shows how far something has come.

watt -- A unit for measuring electrical power, like how much energy a light bulb uses.

backup -- A spare kept ready to use if the main one stops working.

Background you need

Voyager 1 is a spacecraft that left Earth in 1977 and has flown farther than anything else people have built.

It now travels in the space between the stars.

Because it is so far away, even its signals take a very long time to reach us.

Break it down

WHO: NASA's Voyager 1 spacecraft and its engineers on Earth
WHAT: It is almost one light-day from Earth and received a power-saving upgrade
WHERE: Far beyond Earth, in the space between the stars
WHY: Its power is slowly running out, so engineers replaced three devices to keep its science tools working longer

Why it matters

Voyager 1 is like a message from Earth traveling between the stars, and it shows how far human-made things can go.

Test what you remember

1. What is Voyager 1 close to reaching?

- (A) One light-day from Earth
- (B) The Sun
- (C) The Moon
- (D) Mars

2. When did Voyager 1 leave Earth?

- (A) 2020
- (B) Last year
- (C) 1977
- (D) 2000

3. What did engineers replace in September?

- (A) All the cameras
- (B) The whole spacecraft
- (C) The antennas on Earth
- (D) Three devices, with new ones that use less electricity

4. What happened to Voyager's signal in October?

- (A) It never stopped
- (B) It got much louder
- (C) It stopped for about two days, then came back
- (D) It turned into music

5. Why did the signal come back?

- (A) A new satellite arrived
- (B) The spacecraft switched to backup engines by itself
- (C) The wind blew on it
- (D) People flew out to fix it

6. Where is Voyager 1 traveling now?

- (A) Under the ocean
- (B) In the space between the stars
- (C) In a museum
- (D) Around the Moon

Answer key (try the quiz first!): 1: A 2: C 3: D 4: C 5: B 6: B

Questions to think about

1. Mission manager Kareem Badaruddin: Said, 'We have to expect the unexpected.'

MY THOUGHTS (at least 20 words):